

Gmail SMTP Configuration Removal

Overview

This document details the removal of Gmail SMTP configuration from the Mindful Champion application to reduce Google Workspace costs.

Business Context

- **Previous Setup:** 7 Google Workspace licenses (\$42/month)
- **New Setup:** 1 Google Workspace license (\$6/month)
- **Cost Savings:** \$36/month (\$432/year)
- **Retained Email:** dean@mindfulchampion.com

Changes Made

1. Email Service Migration

All email functionality now exclusively uses **Resend API** instead of Gmail SMTP (nodemailer).

2. Files Modified

`lib/email.ts`

- **Before:** Used nodemailer with Gmail SMTP configuration
- **After:** Uses Resend API exclusively
- **Functions Updated:**
 - `sendWelcomeEmail()` - Now uses Resend
 - `sendPartnerRequestEmail()` - Now uses Resend
 - `sendEmail()` - Generic email function now uses Resend
 - `sendWarningEmail()` - Now uses Resend
- **Removed:**
 - `getTransporter()` function (nodemailer)
 - Gmail SMTP configuration (host, port, auth)
 - `EMAIL_USER` and `EMAIL_APP_PASSWORD` environment variable dependencies

`lib/media-center/email-service.ts`

- **Before:** Used Resend with Gmail SMTP fallback via nodemailer
- **After:** Uses Resend exclusively, removed Gmail fallback
- **Changes:**
 - Removed import of `sendWelcomeEmail` from `@/lib/email`
 - Removed fallback logic in `sendWelcomeEmail()` method
 - Added proper error handling and logging for failed emails

`app/api/signup/route.ts`

- **Before:** Used MediaCenterEmailService with Gmail fallback via `sendWelcomeEmail` from `lib/email`

- **After:** Uses MediaCenterEmailService (Resend) exclusively
- **Changes:**
 - Removed import of `sendWelcomeEmail` from `@/lib/email`
 - Removed Gmail fallback logic
 - Simplified error handling

`app/api/auth/forgot-password/route.ts`

- **Before:** Imported `sendEmail` from `@/lib/email` (which used Gmail SMTP)
- **After:** Now uses `sendEmail` from `@/lib/email` (which now uses Resend)
- **No code changes needed:** Function signature remained the same, just backend implementation changed

3. Files Deleted

`lib/email/gmail-service.ts`

- Comprehensive Gmail SMTP integration with multiple sender addresses
- Used nodemailer for email delivery
- **No longer needed:** All functionality moved to Resend

`lib/email/gmail-sender.ts`

- Gmail SMTP sender for video analysis emails
- Used nodemailer with Gmail SMTP
- **No longer needed:** All functionality moved to Resend

4. Files Retained (Unchanged)

`lib/email/resend-client.ts`

- Resend API client initialization
- **Status:** Actively used, no changes needed
- **Configuration:** Uses `RESEND_API_KEY` environment variable

Email Addresses in Use

Resend Configuration (Active)

These email addresses are configured in Resend and continue to work:

- `noreply@mindfulchampion.com` - Default sender for all emails
- `dean@mindfulchampion.com` - Reply-to address
- `partners@mindfulchampion.com` - Partner communications (configured in Resend)

Email Types

All emails now sent via Resend:

1. **Welcome Emails** - New user signups
2. **Partner Request Emails** - Player connection requests
3. **Warning Emails** - Moderation notifications
4. **Password Reset Emails** - Forgot password flow
5. **Trial Expiration Emails** - Trial ending notifications
6. **Admin Notifications** - New user notifications

Environment Variables

Removed (No Longer Used)

- `EMAIL_USER` - Gmail account email address
- `EMAIL_APP_PASSWORD` - Gmail app-specific password
- **Action Required:** These can be removed from Vercel environment variables

Required (Must Remain)

- `RESEND_API_KEY` - Resend API key (CRITICAL - must be set in Vercel)
- **Status:** Should already be configured
- **Verify at:** [https://vercel.com/\[project\]/settings/environment-variables](https://vercel.com/[project]/settings/environment-variables)

Testing Requirements

Post-Deployment Verification

After deploying these changes, test the following email flows:

1. User Signup

- [] Welcome email received
- [] Trial information included
- [] From: Mindful Champion noreply@mindfulchampion.com
- [] Reply-To: dean@mindfulchampion.com

2. Password Reset

- [] Forgot password email received
- [] Reset link works
- [] Email formatting correct

3. Partner Requests

- [] Partner request notification received
- [] Links work correctly

4. Trial Expiration

- [] Trial expiration email received
- [] Upgrade links work

How to Test

1. Go to <https://mindfulchampion.com>
2. Create a test account with your email
3. Verify welcome email is received
4. Test password reset flow
5. Check email headers to confirm Resend delivery

Rollback Plan (If Needed)

If Resend fails and Gmail SMTP needs to be restored temporarily:

1. **Restore Gmail service files** from git history:

```
bash
```

```
git checkout HEAD~1 lib/email/gmail-service.ts
```

```
git checkout HEAD~1 lib/email/gmail-sender.ts
```

2. Restore previous email.ts:

```
bash
```

```
git checkout HEAD~1 lib/email.ts
```

3. Restore environment variables:

- Add `MAIL_USER` back to Vercel
- Add `MAIL_APP_PASSWORD` back to Vercel

4. Redeploy

Cost Analysis

Before

- Google Workspace: 7 licenses × \$6/month = \$42/month
- Resend: Free tier (100 emails/day) or paid as needed
- **Total:** \$42/month minimum

After

- Google Workspace: 1 license × \$6/month = \$6/month
- Resend: Free tier (100 emails/day) or paid as needed
- **Total:** \$6/month minimum

Resend Pricing

- **Free Tier:** 100 emails/day, 3,000 emails/month
- **Growth Plan:** \$20/month for 50,000 emails/month
- **Pro Plan:** \$80/month for 300,000 emails/month

Expected Usage: Given typical user signups and notifications, free tier should be sufficient for early stage. Even with paid plan, total cost would be \$26/month (\$6 Google + \$20 Resend) vs. previous \$42/month.

Dependencies

NPM Packages No Longer Used

The following package is no longer needed but kept for now in case of rollback:

- `nodemailer` - Can be removed in future if no other dependencies

NPM Packages Required

- `resend` - Must remain in package.json
- **Version:** Check `package.json` for current version

Monitoring

Email Delivery Monitoring

- **Resend Dashboard:** <https://resend.com/emails>
- **Check:**

- Delivery rates
- Bounce rates
- Failed emails
- API usage

Application Logs

Monitor Vercel logs for:

-  Email sent successfully - Success messages
-  Failed to send email - Error messages
- `RESEND_API_KEY` errors - Configuration issues

Support Contacts

If Email Issues Occur

1. **Check Resend Dashboard:** <https://resend.com/emails>
2. **Check Vercel Logs:** [https://vercel.com/\[project\]/logs](https://vercel.com/[project]/logs)
3. **Verify Environment Variables:** Ensure `RESEND_API_KEY` is set
4. **Contact:** dean@mindfulchampion.com

Resend Support

- **Documentation:** <https://resend.com/docs>
- **API Status:** <https://status.resend.com>
- **Support:** <https://resend.com/support>

Security Notes

Email Authentication

- **SPF:** Configured for @mindfulchampion.com domain
- **DKIM:** Configured via Resend
- **DMARC:** Recommended to configure for better deliverability

API Key Security

- Store `RESEND_API_KEY` in Vercel environment variables only
- Never commit API keys to git
- Rotate API key if compromised

Completion Checklist

- [x] Remove Gmail SMTP from `lib/email.ts`
- [x] Remove Gmail SMTP from `lib/media-center/email-service.ts`
- [x] Remove Gmail fallback from `app/api/signup/route.ts`
- [x] Delete `lib/email/gmail-service.ts`
- [x] Delete `lib/email/gmail-sender.ts`
- [x] Create this documentation
- [] Deploy to Vercel
- [] Test all email flows
- [] Remove `EMAIL_USER` and `EMAIL_APP_PASSWORD` from Vercel (after verification)

- [] Monitor email delivery for 48 hours
- [] Update team on changes

Additional Notes

- All email templates and content remain the same
- Only the delivery mechanism changed (Gmail SMTP → Resend API)
- No changes required to email HTML/text content
- Email addresses and branding unchanged
- User experience should be identical

Date of Changes

- **Date:** December 18, 2025
 - **Developer:** AI Assistant
 - **Requested By:** Dean (User)
 - **Reason:** Cost reduction - Google Workspace license reduction
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This change is part of a cost optimization initiative to reduce Google Workspace licenses from 7 to 1 while maintaining email delivery functionality.